

The same kilowatt-hour costs twice as much at 6pm as it did at 3am. Flexible plants run on fixed schedules anyway.

District cooling plants, desalination trains, EV fleets and data centres all have demand that could genuinely move within the day. Chilled water can be made early and stored; a desalination train can be throttled; a parked car does not care when it charges. The flexibility exists. The scheduling intelligence does not.

THE OPPORTUNITY, MEASURED

Across four years of German market data, the cheapest third of hours in a day average **84 EUR/MWh** while the priciest third average **166 EUR/MWh**. That gap is available to anyone who can move load across it, and it widens as solar capacity grows. Capturing it needs no new power stations and no batteries, only existing equipment running at different times.

PER MWH SHIFTED OUT OF PEAK HOURS

81 €

cost saved, measured within-day
across 2022–2025

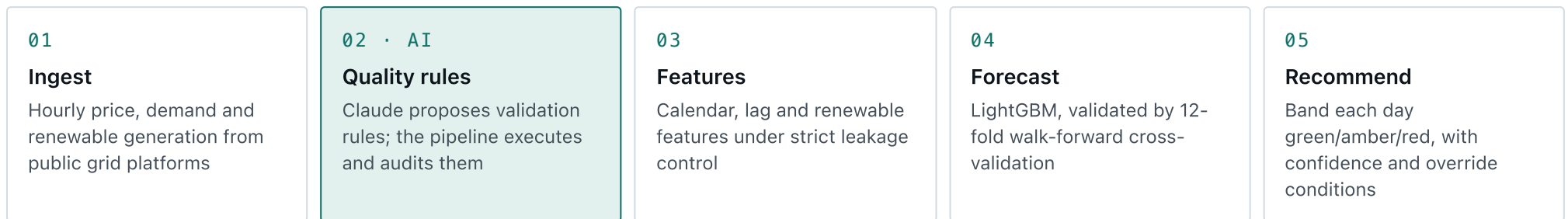
0.034 t

CO₂ avoided, a real but modest co-
benefit

Forecast tomorrow's hourly prices. Tell operators which hours to run in.

GridShift ranks each day's 24 hours into green, amber and red bands, attaches a confidence level, and states the conditions under which its own advice should be ignored. Banding happens **within each day**, because that is the only shift an operator can actually make. You can move a chiller run from 19:00 to 03:00, but you cannot move June demand into April.

PIPELINE



WHO BENEFITS

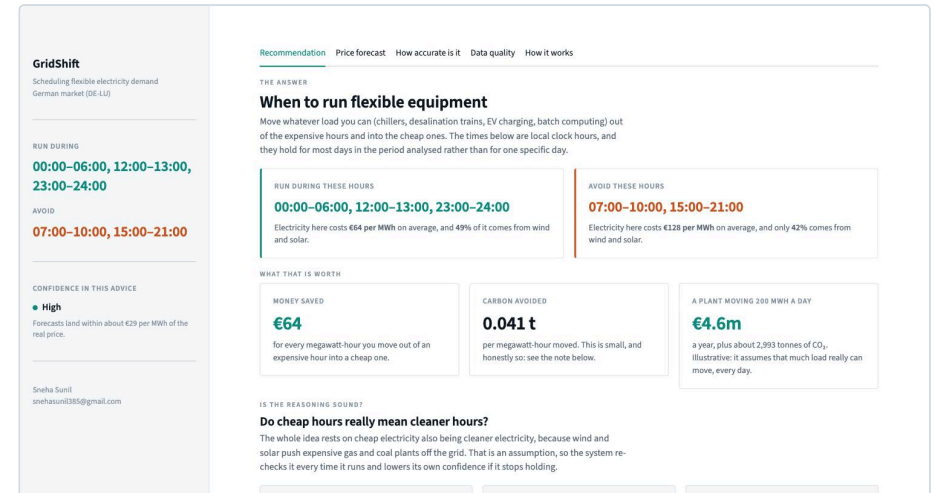
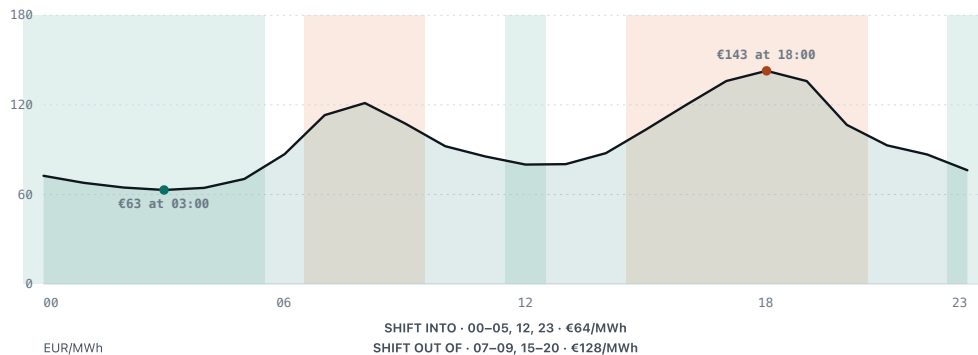
- **District cooling utilities:** chilled water made early and stored
- **Desalination and water authorities:** throttle trains to cheap hours
- **EV charging networks:** shift charging within the parked window
- **Data centre operators:** move batch and training workloads
- **Grid operators:** evening peak shaved, capital investment deferred

BUILT WITH

- **Python, LightGBM:** gradient-boosted hourly price forecasting
- **Claude (claude-haiku-4-5):** automated data-quality rule generation
- **SMARD, ENTSO-E:** four years of free public hourly market data
- **Streamlit:** operator dashboard over the live outputs
- **All data public:** no licensing cost, no procurement barrier to a pilot

Forecast hourly prices, banded into a schedule.

Average forecast price by hour across the Oct–Dec 2025 hold-out window, covering 2,208 hours the model never saw. Green is where GridShift says to run; ember is what it says to avoid. The windows were not hand-picked; they are what the model produced.



The running prototype. It opens with the instruction an operator can act on, in clock hours rather than statistics, and the reasoning behind it sits underneath.

28.8

EUR/MWh forecast error (MAE) on 2,208 unseen hours, **41.8% better** than the seasonal-naive industry baseline of 49.4

12

walk-forward CV folds, expanding window, MAE 23.9 ± 7.3, with zero temporal leakage by construction

70%

empirical coverage of the 80% prediction interval. Under-covered in a volatile quarter, and reported rather than hidden

My headline claim did not survive complete data. That is the most useful thing I found.

I set out to prove something stronger: that price is a good proxy for *carbon intensity*, because abundant wind and solar push fossil plants out of the merit order. If true, a price forecast would double as a forecast of when the grid is clean. An early run gave me **r = 0.86** and I nearly built the whole pitch on it.

THE CORRECTION

That figure was an artifact of a gap in the source data. With wind onshore missing, renewable share was dominated by **solar**, which is strongly diurnal and therefore strongly anti-correlated with price. Restoring the complete wind series cut the correlation to **r = 0.40**. I am reporting the corrected number because the first one would not have survived scrutiny.

AND DECOMPOSING IT MATTERS MORE THAN THE HEADLINE

WHERE THE SIGNAL LIVES	CO ₂ GAP	SHIFTABLE?
Between days (windy vs calm)	0.138 t/MWh	No
Within a day (what you can act on)	0.034 t/MWh	Yes

Most of the price/carbon relationship is between days, but load can only move within one. So the carbon benefit is real and modest, not a silver bullet.

WHAT THIS CHANGES ABOUT THE PITCH

- **Sell it on cost and peak shaving:** 81 EUR/MWh is the strong, defensible claim
- **Treat carbon as a co-benefit:** 0.034 tCO₂/MWh, reported honestly
- **The premise is re-tested every run:** below r = 0.25 the system suspends itself rather than issuing advice it cannot justify
- **The source data is unreliable:** repeated fetches returned different coverage each time; one lost the load series, another returned a 404. Production needs pinned, versioned datasets and coverage assertions *before* modelling

What it is worth, and the one thing that would stop it working here.

BUSINESS VALUE

At the hold-out window's spread of **64 EUR/MWh**, an operator with **200 MWh/day** of genuinely shiftable load would save roughly **4.6M EUR** and **3,000 tCO₂** a year. The four-year average spread is wider, at 81 EUR/MWh; I have quoted the narrower figure because it comes from the window the model was actually tested on. Illustrative either way, since it assumes flexibility every day, which operational constraints would not sustain. The per-MWh figure is the credible claim; the annual number scales with real flexibility.

WHAT DEPLOYMENT ACTUALLY COSTS

Not nothing. The model has to be integrated with the site's control system, and the operator needs interval metering. Two prerequisites matter more than either: **the flexibility must physically exist**, since a cooling plant with no thermal store cannot shift its load at any price, and **the operator must be exposed to time-varying prices**, because on a flat tariff the saving is zero however good the forecast. What GridShift avoids is new generating or storage plant, not integration work.

WHY THAT SUITS PRESIGHT

The customer is a **national utility** such as EWEC, TRANSCO or DEWA rather than a private site, which is Presight's actual customer base. District cooling is the largest single component of Gulf summer peak demand and desalination is a genuinely flexible industrial load; deferring peak generating capacity is a national-scale outcome, not a site-level saving. As solar capacity grows under Energy Strategy 2050, the intraday spread this exploits widens rather than narrows.

THE HONEST TRANSFER PROBLEM

This model consumes a wholesale hourly clearing price. The UAE does not produce one. **EWEC operates as a single buyer** procuring through long-term PPAs, DEWA is vertically integrated, and tariffs are regulated. Lifting this system to Abu Dhabi unchanged would fail, because the input signal does not exist.

The fix makes it a better fit, not a worse one. In a single-buyer market the utility already knows its own dispatch stack, so you forecast **system marginal cost and grid carbon intensity directly** rather than inferring them from price. Everything downstream (the forecasting, the within-day banding, the premise check, the override logic) is unchanged. Only the input series changes.

HOW I WOULD MEASURE SUCCESS

- **Accuracy:** beat the naive baseline by >15% on a 30-day rolling basis
- **Premise integrity:** correlation re-tested every run; below 0.25 the system stops
- **Realised shift:** flexible load actually moved, versus a pre-deployment baseline
- **Cost per MWh consumed:** the number the finance team already tracks
- **Trust:** acceptance and override rates. A system operators ignore has failed, however accurate

Where AI helped, and where I had to stop trusting it.

I used **Claude** (Anthropic), through Claude Code, rather than Copilot. The handout permits other tools, and using the same model the product depends on meant its failure modes were something I had to understand rather than assume.

HOW AI SUPPORTED THIS SUBMISSION

Research the problem	European power market structure, the merit-order effect, and UAE market arrangements. I verified the market-structure claims against primary sources before relying on them.
Generate and refine ideas	Reframing the work from price forecasting to demand scheduling, and pushing back when the sustainability angle stopped being supported by the data.
Structure the proposal	Checking my draft against this handout's required sections and flagging that target users, business value and success measures were missing.
Create visuals and diagrams	The price chart and pipeline diagram, drawn from real pipeline output rather than illustrative numbers.
Improve writing	Editing for concision. I rewrote anything that overstated a result.
Test and validate	Code review, which found both errors described opposite. Every number here comes from code that runs.

AI INSIDE THE PRODUCT ITSELF

Claude reads the data schema and writes 8–12 executable quality rules covering physical limits, market realism and temporal consistency, which the pipeline runs and logs in full. A day of analyst work in seconds, and it generalises: point it at another country and it writes that country's rules.

TWO THINGS AI GOT WRONG, AND WHAT I CHANGED

1. It wrote rules containing a Python operator-precedence bug, where `a <= b | c` parses as `a <= (b | c)`, reporting **720 false data failures** against a true count of **6**. I fixed the prompt and added a guard that quarantines any rule flagging over half the dataset. The guard is the load-bearing fix: it does not depend on the model complying.
 2. My headline correlation of $r = 0.86$ was an artifact of missing wind data. Review caught it; the corrected figure is 0.40, and it changed the entire pitch.
- Both corrections are in this deck rather than hidden. That is the disclosure that matters.**